
Virtual Internship Completion Report

Intelligent Cognitive Alarm Platform

Details

Batch Number: 02

Start Date: 15 July,2026

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Mentor: Alekhya Reddy

Internship Duration: 8 Weeks

1. Project Title

Intelligent Cognitive Alarm Platform

2. Project Objective

The objective of the AI Cognitive Alarm Platform is to develop an intelligent alarm system that helps users wake up consistently and improve their daily habits.

The system uses cognitive challenges, behavioral analytics, habit scoring, notifications, and personalized recommendations to make the alarm experience more effective than a conventional alarm.

Main objectives:

- Provide customizable and recurring alarms.
- Require users to complete cognitive challenges to dismiss alarms.
- Track wake-up, snooze, challenge, sleep, and productivity behavior.
- Calculate a Habit Score based on user behavior.
- Provide personalized recommendations and reminders.
- Provide separate dashboards for User, Coach, and Admin.
- Generate analytics and reports for monitoring user progress.

3. Project Description in Detail

The AI Cognitive Alarm Platform is a web-based intelligent alarm management and behavioral analytics system.

Users can create and manage multiple alarms with options such as daily, weekday, weekend, one-time, and smart adaptive alarms. When an alarm is triggered, the user can be required to complete a cognitive challenge such as a math problem, logic puzzle, memory challenge, word game, pattern recognition, riddle, or quiz.

The platform records user interactions such as wake-up verification, snoozing, challenge performance, and behavioral events. This data is analyzed to understand the user's habits and calculate a Habit Score.

The system provides different dashboards:

- **User Dashboard:** Alarm management, challenges, notifications, wellness information, analytics, and personal reports.
- **Coach Dashboard:** User behavior insights, habit adherence, sleep trends, and progress monitoring.
- **Admin Dashboard:** User management, platform analytics, system information, and platform announcements.

The platform also provides notifications and reminders, including bedtime reminders, wake-up reminders, habit alerts, challenge reminders, and progress notifications.

Reports can be generated for habit performance, wake-up behavior, challenge performance, productivity, and sleep analytics, with PDF and Excel export options.

3. Approach Used

The project was developed using a modular full-stack approach.

1. **Requirement Analysis**
Identified requirements for alarms, cognitive challenges, analytics, notifications, dashboards, reports, and user roles.
2. **Frontend Development**
Developed responsive User, Coach, Admin, Login, and authentication interfaces using HTML, CSS, and JavaScript.
3. **Backend Development**
Built REST APIs using Node.js and Express.js for authentication, alarms, challenges, analytics, notifications, reports, and user management.
4. **Database Integration**
Used PostgreSQL to store users, alarms, behavioral events, challenge performance, notifications, and wake-up verification data.
5. **Authentication & Security**

Implemented JWT-based authentication, role-based access control, password security, and Google OAuth authentication.

6. **Analytics & Personalization**

Analyzed behavioral data to calculate habit scores, performance metrics, trends, and personalized recommendations.

7. **Testing & Deployment**

APIs were tested using Postman, frontend/backend integration was tested, and the application was deployed using **Vercel with Neon PostgreSQL**

4. Technologies Used

- **Frontend**

The web frontend was developed using HTML5, CSS3, and JavaScript. JavaScript was used to implement user interactions, authentication handling, alarm management, notifications, cognitive challenges, and communication with the backend REST APIs. **Chart.js** was used to create analytics visualizations for behavioral data, habit scores, challenge performance, sleep trends, and productivity.

- **Backend**

The backend was implemented using Node.js and Express.js to provide RESTful APIs for authentication, alarm management, cognitive challenges, behavioral analytics, notifications, recommendations, and reports. JWT-based authentication was used to secure user accounts, with Google OAuth using Passport.js for Google login. Role-based access control was implemented for User, Wellness Coach, and Administrator roles.

- **Database**

PostgreSQL was used as the primary relational database for storing structured application data such as user accounts, alarms, notifications, behavioral events, challenge performance, and wake-up verification records. Neon PostgreSQL was used as the cloud database for the deployed application.

- **AI, Behavioral Analytics and Adaptive Intelligence**

The platform uses behavioral analytics and rule-based intelligence to analyze wake-up behavior, snooze patterns, challenge performance, sleep-related activity, productivity, and habit consistency. A Habit Score is calculated using wake-up consistency, challenge completion, snooze reduction, and sleep schedule adherence. The system also provides personalized recommendations and adaptive alarm functionality based on user behavior. Google Gemini API was integrated for AI-based functionality.

- **Notifications and Reports**

The platform includes a notification system for bedtime reminders, wake-up reminders, habit alerts, challenge reminders, progress notifications, and platform announcements. Reports are generated for habit performance, wake-up behavior, challenge performance, productivity, and sleep analytics, with PDF and Excel export functionality.

- **Other Libraries, Tools and Deployment**

Chart.js was used for dashboard visualizations. Postman was used for REST API testing, while Visual Studio Code was used for development. Git and GitHub were used for version control. The frontend and backend were deployed using Vercel, with Neon PostgreSQL used as the production database.

5. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project induction, environment setup, and requirement understanding.	Attended induction session, understood project requirements, and set up the frontend and backend development environment.
Week 2	Design system architecture, database schema, and implement authentication. (Milestone 1 Review)	Designed the system architecture and database schema, implemented JWT-based authentication with role-based access, and built the alarm scheduling module.
Week 3	Build the cognitive challenge engine and puzzle generation workflows.	Implemented the cognitive challenge engine supporting math problems, logic puzzles, memory challenges, and riddles, with challenge validation logic.
Week 4	Develop wake-up verification and challenge evaluation system. (Milestone 2 Review)	Built the wake-up verification module with multistep and time-based challenge checks, and integrated challenge performance tracking.
Week 5	Implement the adaptive difficulty engine and behavioral analytics.	Developed the adaptive difficulty engine that adjusts challenge complexity based on user performance, and began behavioral analytics tracking (snooze patterns, wake-up trends).
Week 6	Build habit scoring model and recommendation engine. (Milestone 3 Review)	Implemented the weighted habit scoring model and the recommendation engine for personalized sleep and productivity suggestions.
Week 7	Build dashboards, notifications, and reports; UI verification and testing.	Developed user, wellness-coach, and admin dashboards, added the notification and reminder system, and carried out UI verification and functional testing.
Week 8	Final integration, testing, deployment, and documentation.	Completed end-to-end integration testing, deployed the application on Vercel, and finalized the presentation and internship report.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Attended the kick-off session, understood project requirements, and completed the initial project setup.	17 July
Prototype / First Draft	Implemented authentication, alarm scheduling, and the initial cognitive challenge engine.	4 August
Mid-Term Review	Completed the wake-up verification module, adaptive difficulty engine, and behavioral analytics workflows.	14 August
Final Submission	Developed the frontend and backend, implemented habit scoring, the recommendation engine, and dashboards; deployed the application on Vercel.	28 August
Presentation	Prepared the project presentation, demonstrated the application, and explained the implementation and outcomes.	14 Sept

5b. Project Execution Details

• Requirement Analysis

Development began by identifying the main problems with conventional alarms, such as easy snoozing, inconsistent wake-up behavior, and lack of behavioral feedback. Based on these requirements, modules for **alarm management, cognitive challenges, wake-up verification, behavioral analytics, habit scoring, notifications, recommendations, dashboards, and reports** were defined and developed separately.

• System Design

The platform follows a **client-server architecture** consisting of a web-based frontend, a Node.js and Express.js backend, and a PostgreSQL database. The frontend communicates with the backend through RESTful APIs, while the database stores user, alarm, behavioral, challenge, notification, and wake-up verification data.

• Authentication and Access Control

Authentication was implemented using **JWT-based authentication**, with **Google OAuth** support for Google login. Role-based access control was implemented for the **User, Wellness Coach, and Administrator** roles. Protected APIs verify the user's authentication token and role before providing access to restricted resources.

• Alarm Management

The Alarm Management Module supports **multiple alarms** with different scheduling options, including Daily, Weekday, Weekend, One-Time, and Smart Adaptive alarms. Users can customize alarm labels,

sounds, vibration, snooze settings, and challenge difficulty. Alarm data is stored in PostgreSQL and retrieved dynamically through backend APIs.

- **Cognitive Challenge and Wake-Up Verification**

The Cognitive Challenge Module provides different challenge types, including **math problems, logic puzzles, memory challenges, word games, pattern recognition, riddles, and quick quizzes**. Five difficulty levels are supported: **Beginner, Easy, Moderate, Hard, and Expert**. Challenge performance and wake-up verification are recorded to analyze user engagement and wake-up behavior.

- **Behavioral Analytics**

The Behavioral Analytics Module analyzes **snooze patterns, wake-up behavior, challenge performance, sleep-related activity, productivity, and habit consistency**. The collected data is displayed through interactive charts and analytics on the User and Coach dashboards.

- **Habit Scoring and Recommendations**

The Habit Scoring Engine calculates a user's overall habit score using four factors: **wake-up consistency (35%), challenge completion (25%), snooze reduction (20%), and sleep schedule adherence (20%)**. The score and behavioral data are used to provide recommendations related to improving wake-up habits, sleep schedules, and productivity.

- **Notifications and Reports**

A notification system provides **bedtime reminders, wake-up reminders, habit alerts, challenge reminders, progress notifications, and platform announcements**. The reporting module provides habit, wake-up, challenge performance, productivity, and sleep analytics, with **PDF and Excel export** functionality.

- **Role-Based Dashboards**

Three dashboards were developed according to user roles. The **User Dashboard** provides alarm management, challenges, notifications, habit score, wellness analytics, and personal reports. The **Wellness Coach Dashboard** provides behavior insights, habit adherence, sleep trends, and progress monitoring. The **Admin Dashboard** provides user management, platform analytics, system information, and platform announcements.

- **Testing and Deployment**

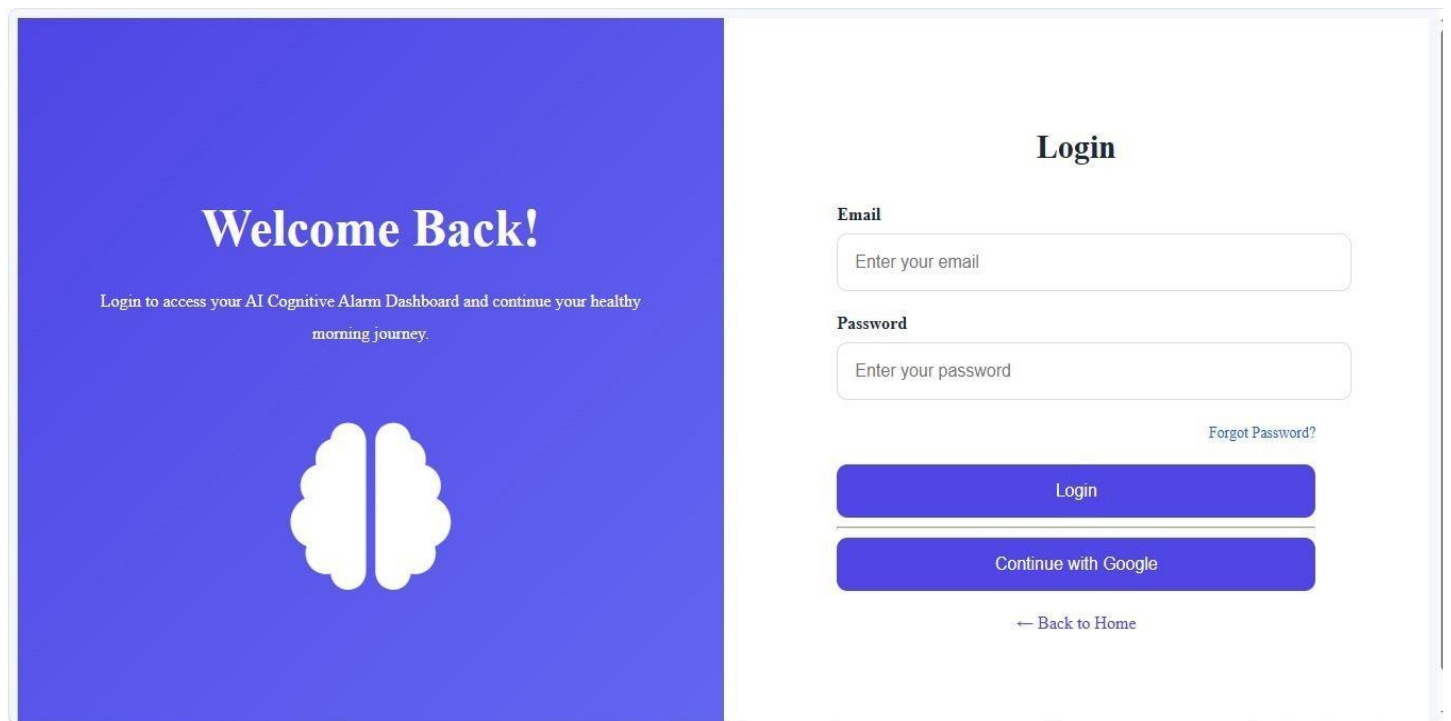
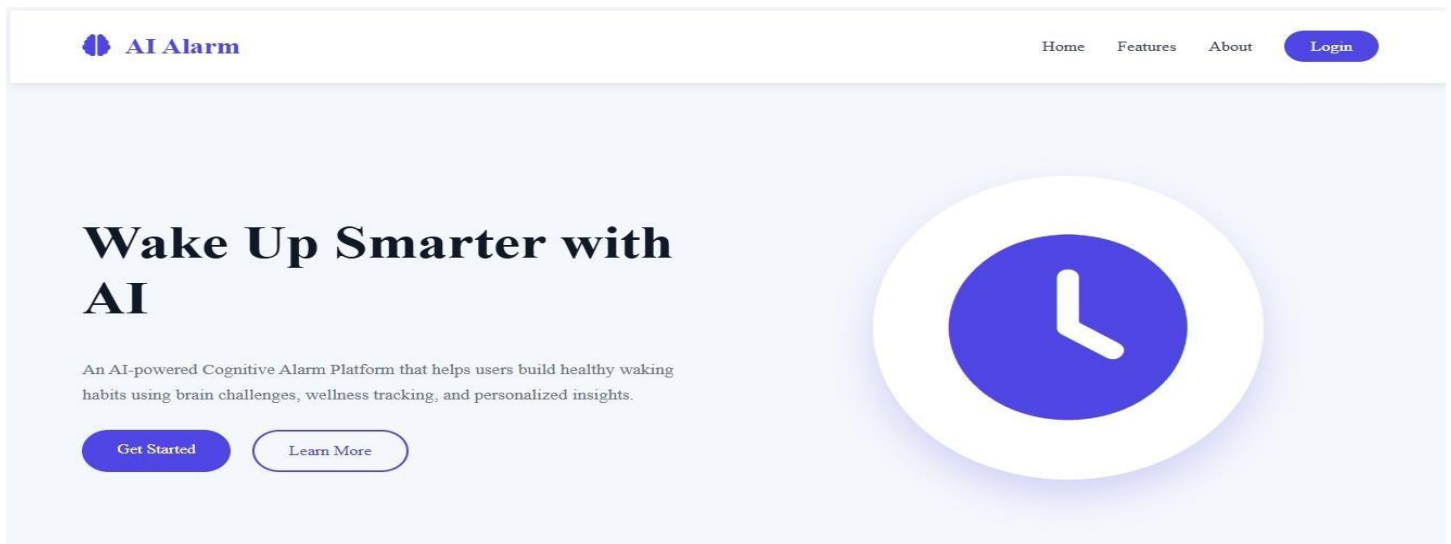
The system was tested using **Postman for API testing** and browser-based testing for frontend functionality. Authentication, API authorization, alarm operations, challenge generation, challenge performance, analytics, notifications, reports, and dashboard functionality were tested. The application was deployed using **Vercel**, with **Neon PostgreSQL** used as the production database.

- **Final Outcome**

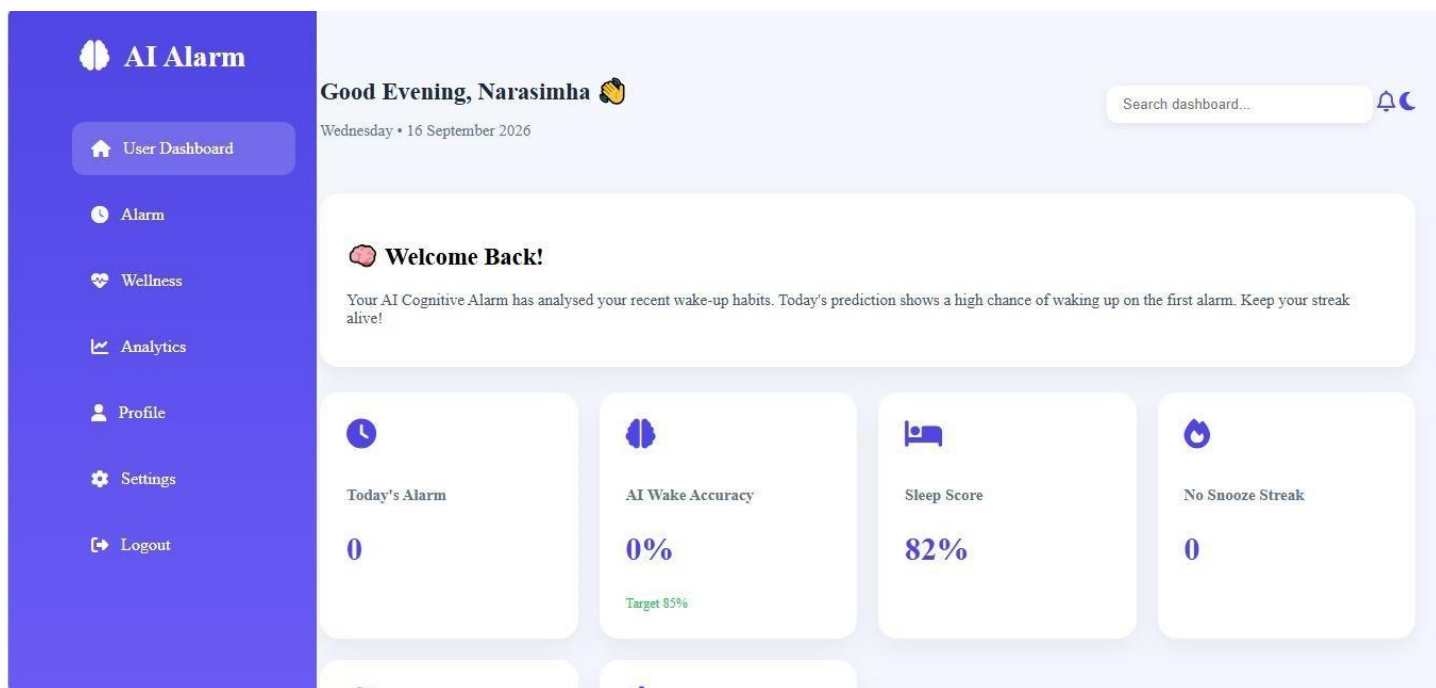
The completed project resulted in a functional **AI Cognitive Alarm Platform** that combines intelligent alarm management, cognitive challenges, wake-up verification, behavioral analytics, habit scoring, personalized recommendations, notifications, reports, and role-specific dashboards in a single web application.

6. Snapshots / Screenshots

Landing and signup page



User Dashboard



Analytics



Virtual Internship Portal

Admin

Admin Panel

Dashboard
Admin Dashboard
Users
Coaches
Analytics
Reports
System
Platform Announcements
Settings
Logout

Administrator Dashboard

AI Cognitive Alarm Platform Control Center

Total Users

11

Total Coaches

3

Challenge Accuracy

97%

Platform Health

88%

Today's Alarms

0

Average Wakefulness

7.82/10

Platform Announcements

Send an announcement to all registered users.

Announcement Title

Coach dashboard

Coach Panel

Coach Dashboard
Assigned Users
Logout

Welcome Coach

Monitor user wellness and alarm performance

Assigned Users

11

Healthy Sleep

4

Missed Alarms

0

Average Accuracy

54%

User Behavior Insights

Metric	Value
Average Wake-up Success	47%
Average Productivity	51%

7. Challenges Faced

- **Cognitive Challenge Generation:** Designing different types of challenges such as math problems, logic puzzles, memory challenges, word games, pattern recognition, riddles, and quick quizzes across five difficulty levels.
- **Challenge Difficulty Management:** Ensuring that the challenge difficulty was appropriate for the user's performance and supported levels from Beginner to Expert.
- **Wake-Up Verification:** Implementing challenge-based wake-up verification so that the alarm could record whether the user successfully completed the required challenge.
- **Behavioral Analytics:** Collecting and processing snooze patterns, wake-up behavior, challenge performance, productivity, and sleep-related data to generate meaningful analytics.
- **Habit Score Calculation:** Implementing the weighted Habit Score using wake-up consistency, challenge completion, snooze reduction, and sleep schedule adherence while ensuring the dashboard values were updated correctly.
- **Authentication & Security:** Implementing JWT authentication, Google OAuth, and role-based access control for User, Wellness Coach, and Administrator roles. Protecting APIs from unauthorized access was also an important challenge.
- **Frontend–Backend Integration:** Ensuring that the HTML, CSS, and JavaScript frontend communicated correctly with the Node.js/Express backend APIs and that dynamic data was displayed after login and page refresh.
- **Dashboard Data Consistency:** Resolving issues where analytics and dashboard values initially displayed incorrect or zero values after refresh due to authentication tokens and user ID handling.
- **Notifications and Reports:** Implementing reminders, habit alerts, progress notifications, platform announcements, and dynamic PDF/Excel reports while ensuring that the correct user data was displayed.
- **Deployment & Database Integration:** Configuring the application for production deployment using Vercel and Neon PostgreSQL, including database connectivity, SSL configuration, environment variables, and Google OAuth redirect URLs.

8. Learnings & Skills Acquired

- Developed practical understanding of behavioral analytics, habit scoring, cognitive challenges, and intelligent alarm systems.
- Gained hands-on experience with HTML, CSS, JavaScript, Node.js, Express.js, and PostgreSQL for full-stack web application development.

- Learned to design and implement RESTful APIs for authentication, alarms, challenges, analytics, notifications, recommendations, and reports.
- Acquired practical knowledge of JWT authentication, Google OAuth, and role-based access control for User, Wellness Coach, and Administrator roles.
- Learned to implement cognitive challenge generation and performance tracking across multiple challenge types and difficulty levels.
- Improved understanding of weighted scoring models and recommendation logic using behavioral data to calculate Habit Scores and provide personalized suggestions.
- Gained experience in building interactive dashboards and data visualizations using Chart.js, including habit, snooze, wake-up, sleep, challenge, and productivity analytics.
- Learned to implement notifications, PDF/Excel report generation, API testing with Postman, and database integration with Neon PostgreSQL.
- Enhanced debugging, frontend-backend integration, security testing, Git/GitHub version control, and Vercel deployment skills.
- Strengthened problem-solving, documentation, communication, teamwork, and project management skills through the development, testing, and deployment of the project.

9. Testimonial

- This internship provided me with valuable hands-on experience in developing and deploying a real-world webbased AI Cognitive Alarm Platform. Working on the project helped me understand how the frontend, backend, database, authentication, behavioral analytics, and intelligent features work together to solve a practical problem.
- Implementing the alarm management, cognitive challenge system, wake-up verification, habit scoring, notifications, personalized recommendations, dashboards, and reporting modules helped me improve my technical knowledge and problem-solving skills. I also gained practical experience with Node.js, Express.js, PostgreSQL, JavaScript, REST APIs, JWT authentication, Google OAuth, Chart.js, Git/GitHub, Postman, and Vercel deployment.
- The development process improved my ability to debug issues, integrate frontend and backend components, test APIs, manage project tasks, and document technical work. Overall, this internship strengthened both my technical and professional skills and gave me practical experience in developing, testing, and deploying a complete full-stack application.

10. Conclusion

- This internship provided me with valuable practical experience in full-stack web application development and AI-based application development. I gained hands-on experience in developing the AI Cognitive Alarm Platform using technologies such as HTML, CSS, JavaScript, Node.js, Express.js, PostgreSQL, Chart.js, and Google Gemini API.
- Throughout the project, I strengthened my skills in REST API development, database management, authentication, behavioral analytics, cognitive challenge implementation, notifications, reporting, and frontend-backend integration. I also gained practical experience in JWT authentication, Google OAuth, role-based access control, API testing with Postman, version control using Git and GitHub, and deployment using Vercel and Neon PostgreSQL.
- The internship helped me bridge the gap between academic learning and practical industry development by allowing me to design, develop, test, debug, and deploy a complete real-world application. It also improved my problem-solving, debugging, communication, documentation, and project management skills.
- Overall, the experience strengthened my confidence in developing intelligent software solutions and encouraged me to further pursue Artificial Intelligence, Machine Learning, and Full-Stack Development.

11. Acknowledgement

I would like to express my sincere gratitude for the opportunity to participate in this Virtual Internship program. It offered valuable exposure to real-world software development and Artificial Intelligence, enabling me to apply my academic knowledge to a practical, industry-oriented project.

I would like to extend my heartfelt thanks to my Project Mentor, Ms. Alekhya Reddy, for her continuous guidance, encouragement, and valuable feedback throughout the internship. Her support and technical insights played a significant role in the successful completion of this project.

My sincere thanks also go to the team members and coordinators for their cooperation, teamwork, knowledge sharing, and continuous support during the development of the project. Their support helped me successfully complete this internship and further strengthened my technical knowledge, professional skills, and confidence in pursuing a career in Artificial Intelligence and Software Development.